Phase 1: Foundations (Week 1)

Objective: Understand database design, normalization, and Python-MySQL integration basics.

1. What You'll Learn

- Fundamentals of Relational Database Design.
- How to normalize a database schema for efficiency and scalability.
- Setting up a local development environment.

2. Hands-On Tasks

- Install and set up MySQL.
- Design and normalize the database schema for ProFarmer.
- Write Python scripts to connect to MySQL, create tables, and insert basic data.

Key Milestone by End of Week 1:

- ProFarmer database schema fully implemented in MySQL.
- Python script to add farmers and retrieve data.

Phase 2: Functionalities (Weeks 2-3)

Objective: Build CRUD functionalities (Create, Read, Update, Delete) for each module in the database.

1. What You'll Learn

- Writing efficient SQL queries.
- Python-based user interfaces (CLI first, then GUI).
- Error handling and debugging.

2. Hands-On Tasks

- Implement CRUD operations for farmers, farms, crops, and related tables.
- Build a basic Tkinter GUI to interact with the database.
- Test database relationships using foreign keys.

Key Milestone by End of Week 3:

- Fully functional GUI for managing farmers and farms.
- Database filled with sample data for testing.

Phase 3: Advanced Features (Weeks 4–5)

Objective: Enhance the system with analytics, automation, and usability.

1. What You'll Learn

- Writing complex queries for reporting and analytics.
- Using Python libraries like Pandas and Matplotlib for data visualization.
- Adding notifications or alerts (e.g., low stock, market updates).
- User authentication and role-based access.

2. Hands-On Tasks

- Generate insights (e.g., total yields, expenses, revenue) using SQL and Python.
- Create visual dashboards with graphs and charts.
- Implement search and filter functionalities in the GUI.

Key Milestone by End of Week 5:

- ProFarmer system provides actionable insights.
- GUI includes analytics and visualizations.